

Genome Engineering Techniques

Purpose: Make precise, targeted changes (insert, delete, modify, replace DNA) directly in a cell's genome.

1. Key Genome Editing Tools

Tool	How it Works	Special Feature
CRISPR-Cas9	Uses RNA guide to direct Cas9 enzyme to a specific DNA site, cutting it for repair with desired changes	Most widely used; simple and efficient
TALENs	Engineered proteins bind to DNA and create double-strand breaks	Customizable for specific sequences
ZFNs	Artificial proteins target and cut DNA, triggering repair	Similar to TALENs
Meganucleases	Natural enzymes that recognize long DNA sequences and cut	Can be engineered for targeted editing
Base Editing	Chemically converts one DNA base to another without cutting the DNA backbone	Precise point mutation editing
Gene Silencing / Knockdown (RNAi)	Short RNA molecules block expression of specific genes without changing DNA	Useful to reduce gene activity temporarily

<https://www.youtube.com/watch?v=U3Z4uODKbxQ>

2. General Workflow

1. **DNA Extraction / Isolation** – Obtain genetic material from the organism.
2. **Gene Cloning / Synthesis** – Copy the gene of interest.

3. **Gene Design / Modification** – Add promoters or modify sequences for correct function.
4. **Transformation / Delivery** – Insert the gene into target cells.
5. **Selection & Regeneration** – Identify successfully modified cells and grow them into a complete organism (e.g., transgenic plants).

Modern genome editing allows scientists to precisely control genes in organisms for medicine, agriculture, biofuels, or research, using techniques that range from RNA-guided cutting (CRISPR) to base conversion without DNA breaks.

2. DNA Repair Mechanisms Used

- **Homologous Recombination (HR):** Precise insertion or modification using a DNA template.
 - **Non-Homologous End Joining (NHEJ):** Less precise repair that creates small insertions or deletions; used to knock out genes.
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3. Traditional Methods

- **Molecular Cloning / Recombinant DNA:** Copying and combining DNA sequences in the lab.
 - **Gene Silencing:** Reducing or blocking gene expression.
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4. DNA Delivery Methods

- **Microinjection:** Directly injecting DNA into cells (e.g., egg cells).
 - **Electroporation:** Using electric pulses to make cell membranes temporarily permeable.
 - **Viral Vectors:** Modified viruses deliver DNA into cells.
 - **Gene Gun:** Shooting DNA-coated particles into plant cells.
 - **Bacterial Transformation:** Plasmids carrying genes are introduced into bacteria.
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Genome engineering combines precise gene-editing tools, natural repair pathways, and effective DNA delivery methods to modify organisms for research, medicine, agriculture, or industry.

CRISPR-Cas9 Gene Editing

Definition:

CRISPR-Cas9 is a precise gene-editing tool that uses a **guide RNA (gRNA)** to direct the **Cas9 enzyme** to a specific DNA sequence, creating a cut that can be repaired to modify the genome. It is adapted from a natural bacterial defense system.

How CRISPR-Cas9 Works

Step	Description
Targeting	A synthetic gRNA is made to match the DNA sequence of interest. Cas9 binds to this gRNA.
Binding	gRNA guides Cas9 to the DNA target. Cas9 binds only if a nearby PAM sequence is present.
Cutting	Cas9 acts as molecular scissors, cutting both strands of DNA (double-strand break).
Repair	Cell repairs the break using: • Non-Homologous End Joining (NHEJ) : introduces small insertions/deletions, can disrupt gene function. • Homology Directed Repair (HDR) : uses a provided DNA template to make precise edits.

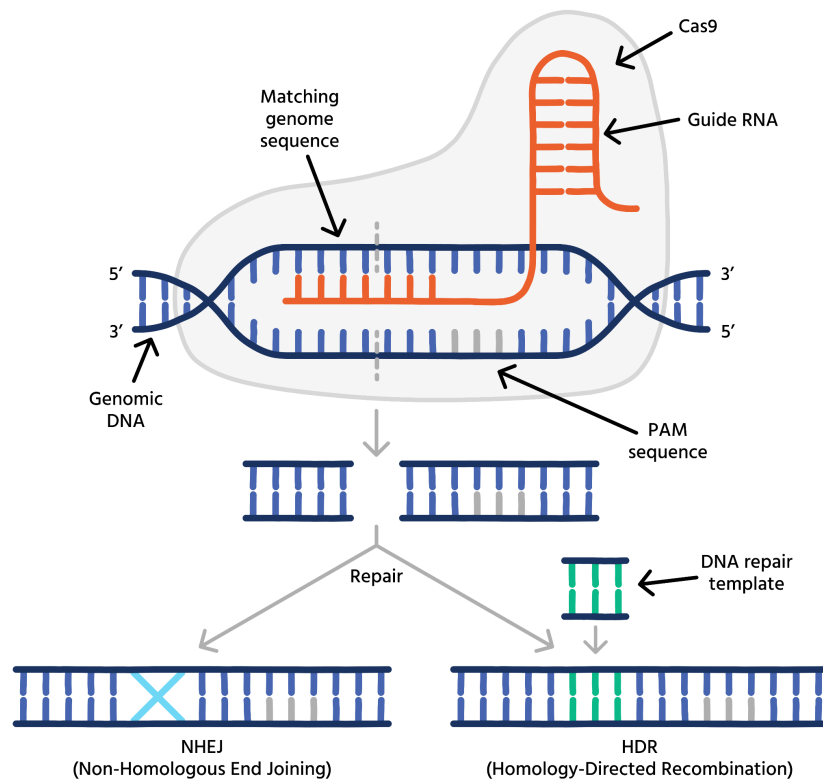
Applications

Area	Use
Scientific Research	Study gene function by modifying genomes of model organisms.
Agriculture	Improve crop and livestock traits.
Medicine	Develop treatments for genetic diseases and infectious diseases (e.g., HIV).

Therapeutics

Correct disease-causing mutations (still under research for clinical safety and efficiency).

Summary: CRISPR-Cas9 allows **targeted, efficient genome editing** by cutting DNA at precise locations and letting natural repair mechanisms make desired changes. Its versatility makes it widely used in research, agriculture, and medicine.



Zinc-Finger Nucleases (ZFNs) for Genome Engineering

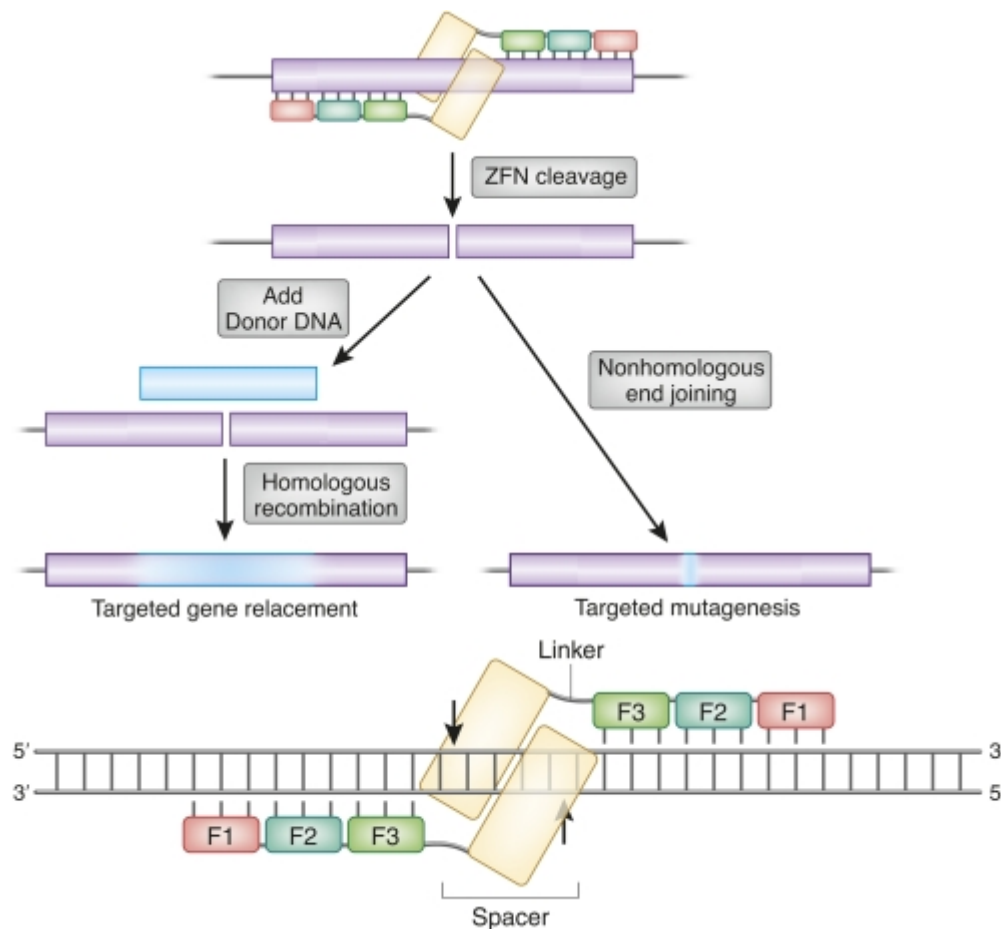
ZFNs are engineered proteins that allow **precise gene targeting** by creating double-strand breaks (DSBs) at specific DNA sites. They consist of two parts: a **DNA-binding domain** made of zinc fingers, each recognizing 3 base pairs, and a **FokI cleavage domain** that cuts DNA when dimerized. Two ZFNs bind neighboring sequences, bringing the FokI domains together to make a targeted cut. This dimerization ensures high specificity, reducing off-target effects.

ZFNs stimulate the cell's natural **DNA repair pathways**:

- **Non-Homologous End Joining (NHEJ):** introduces small insertions/deletions, useful for disrupting genes.
- **Homology-Directed Repair (HDR):** uses a provided DNA template for precise gene replacement or modification.

Historically, gene targeting was limited to organisms like yeast and mice because efficient homologous recombination was rare in most cells. Introducing DSBs dramatically increased targeting efficiency, inspiring the use of ZFNs in a wide range of organisms. Early successes included targeted mutagenesis and gene replacement in *Drosophila*, demonstrating activity in somatic and germline cells.

ZFNs remain a versatile genome-editing tool, achieving modification rates of up to 10% in some cases. They are valuable for research, biotechnology, and potential therapeutic applications, including human gene therapy, though design and optimization of zinc-finger arrays remain technically challenging.



TALENs – Targeted Genome Editing Technology

- **Purpose:**
 - Engineered nucleases for targeted DNA modification in cells and organisms.
 - Can be used for research and potential therapies for genetic diseases.
- **Mechanism:**
 - Composed of:

- **Custom DNA-binding domain** → recognizes specific DNA sequences.

- **Non-specific FokI nuclease domain** → cleaves DNA.

- Induces **double-strand breaks (DSBs)** at targeted sites.
- Cells repair DSBs via endogenous repair pathways → allows precise genome edits (mutations, deletions, insertions, etc.).

- **Comparison with Zinc Finger Nucleases (ZFNs):**

- ZFNs: DNA-binding domain = zinc finger proteins (3 bp per finger).
- TALENs: DNA-binding domain = TALE repeats (1 repeat per base, easier to design).
- Both use FokI nuclease → dimerization required for DNA cleavage.

- **TALE Repeats:**

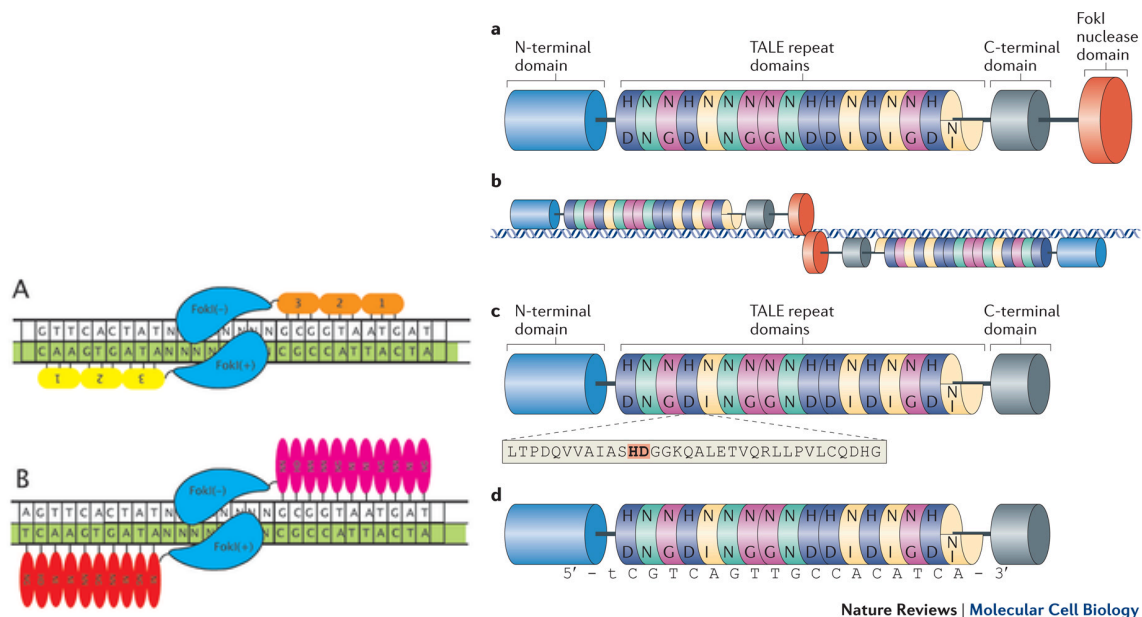
- Highly conserved 33–35 amino acid repeats.
- Two hypervariable residues per repeat determine base recognition (the “TALE code”).
- TALENs bind as dimers → cleavage occurs in the **spacer region** between binding sites.
- Often require a **5’ thymine** before the first TALE repeat binding site.

- **Advantages of TALENs:**

- Easier and more predictable to design than ZFNs.
- High specificity and versatility for targeting almost any DNA sequence.
- Applicable to many cell types and organisms.

- **Applications:**

- Targeted gene knockouts, insertions, or corrections.
- Potential therapeutic genome editing for monogenic diseases (e.g., sickle cell anemia, cystic fibrosis).



RNAi Gene Knockdown Using siRNA – Key Points

- **What RNAi is:**
 - Biological process where dsRNA triggers sequence-specific mRNA degradation.
 - Silences genes post-transcriptionally.
- **siRNA Basics:**
 - 21–23 bp duplexes, generated chemically or via shRNA expression.
 - Incorporated into RISC to target mRNA.
- **Controls:**
 - **Negative:** non-targeting or scrambled siRNA.
 - **Positive:** siRNA with known knockdown effect.
- **siRNA Preparation:**
 - Resuspend lyophilized duplex in RNase-free water (20 µM).

- Anneal single strands if necessary (heat, then slow cool).

- **Delivery Methods:**

- **Lipid-based transfection** (e.g., siLenFect™) – good for adherent cells.
- **Electroporation** (e.g., Nucleofector™) – high efficiency for hard-to-transfect cells; may cause cell death.

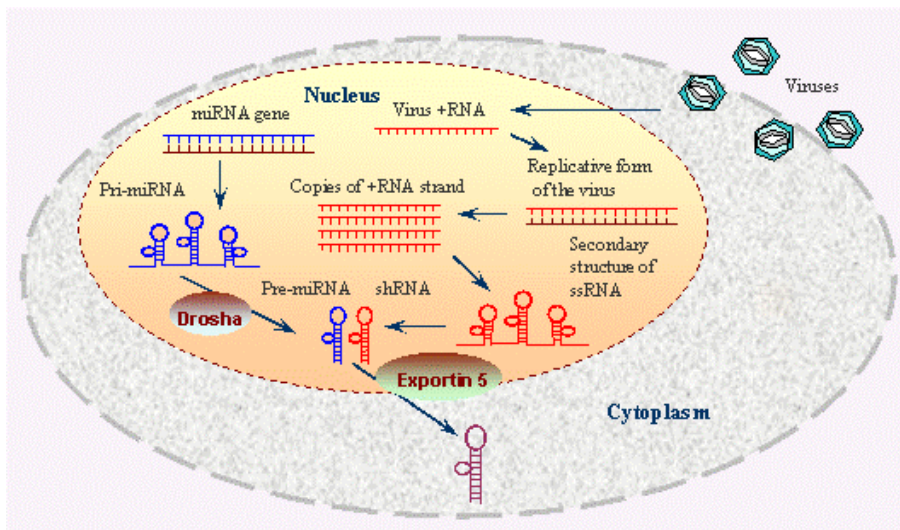
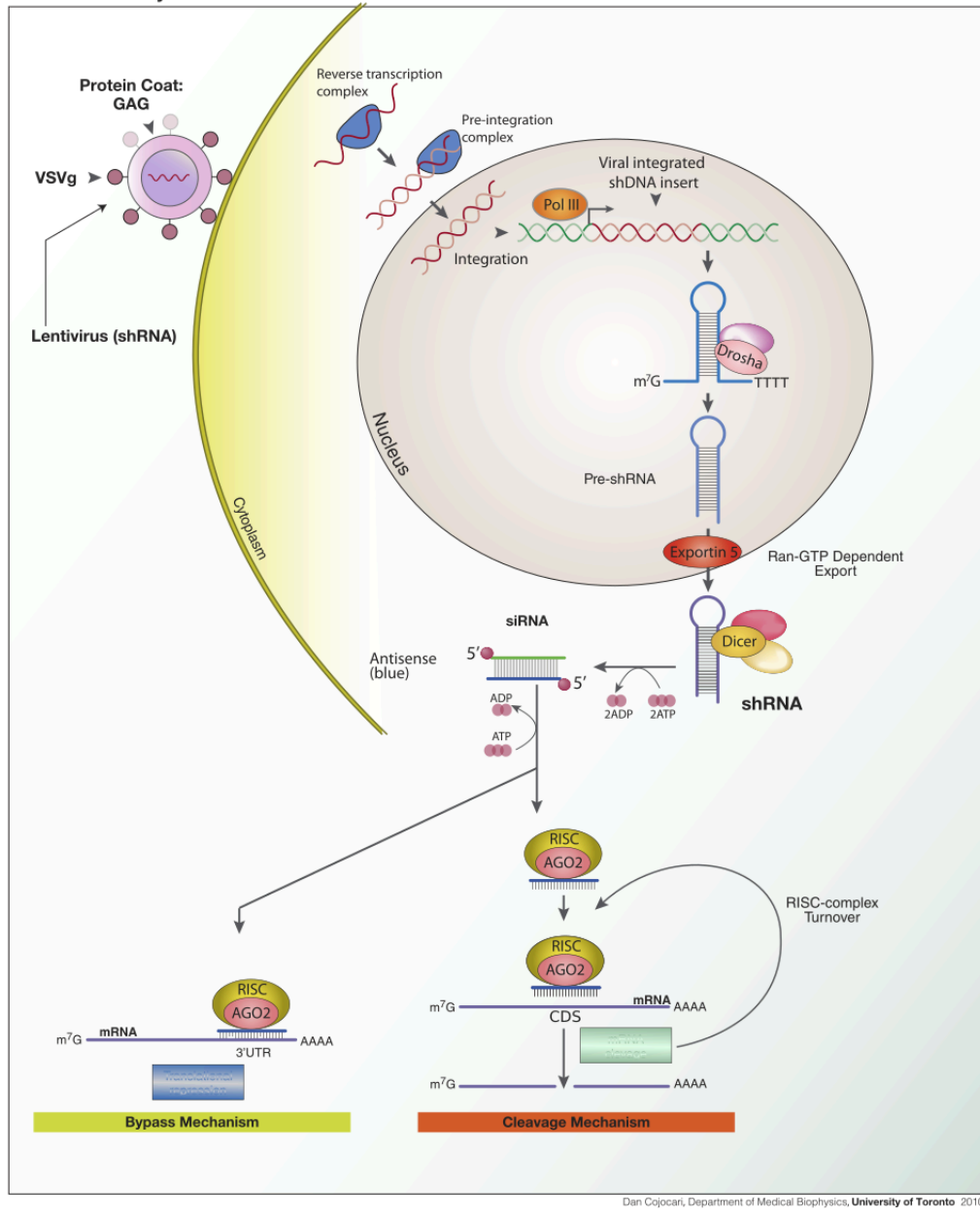
- **Optimization Tips:**

- Test multiple siRNA sequences per gene.
- Adjust siRNA concentration (5–300 nM).
- Select method depending on cell type and downstream assays.

- **Assessment:**

- RT-PCR for mRNA knockdown.
 - Western blot or immunoassays for protein reduction.
 - Functional assays to test effects (growth, apoptosis, signaling).
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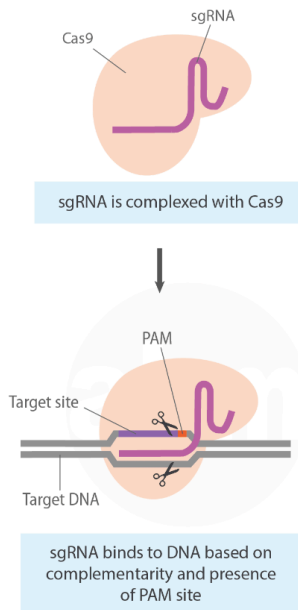
Lentiviral Delivery of shRNAs and the Mechanism of RNAi Interference in Mammalian Cells.



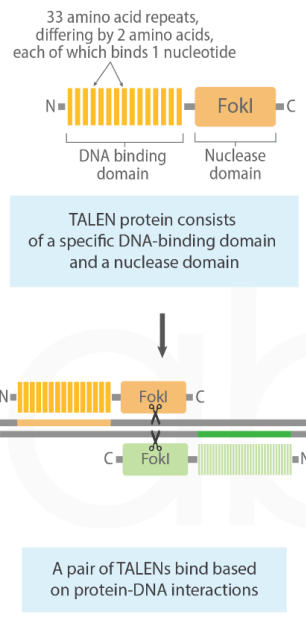
Gene Knockout

Gene expression is eliminated by creating InDels in the DNA

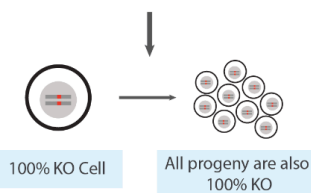
Using CRISPR



Using TALEN



Cas9/TALENs cause a dsDNA break. The process of repair via Non-Homologous End Joining (NHEJ) introduces InDel mutations which disrupt the reading frame of the gene.

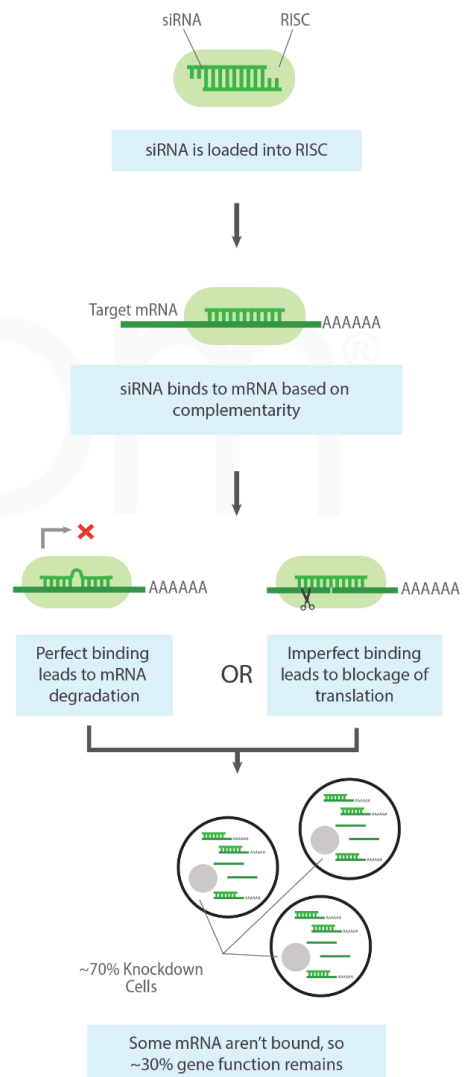


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Gene Knockdown

Gene expression is reduced by targeting the mRNA

Using RNAi



	RNAi	TALE Repression	TALEN	Cas9 Nuclease	CRISPRi	CRISPRa
Loss-of-function mechanism	Post-transcriptional RNA degradation	Repression of transcription	Frame shift DNA mutation	Frame shift DNA mutation	Repression of transcription	Activation of transcription
Result	Reversible knockdown	Reversible knockdown	Permanent knockout	Permanent knockout	Reversible knockdown	Reversible activation
Transgenes	si/shRNA	TALE-KRAB	TALEN	Cas9 nuclease	dCas9-KRAB	dCas9-VP64
				sgRNA	sgRNA	sgRNA
Guiding sequence	si/shRNA	DBD	DBD	sgRNA	sgRNA	sgRNA
Required sequence information	Transcriptome	Annotated TSS	Transcriptome	Transcriptome	Annotated TSS	Annotated TSS
Off-target space	Transcriptome	Window around TSS	Genome; requires FokI dimerization	Genome; cuts as monomer	Window around TSS	Window around TSS
Transcript variants	All variants via conserved region	Only variants from the same TSS	All variants via conserved region	All variants via conserved region	Only variants from the same TSS	Only variants from the same TSS